

Demystify Mixed Model Equation of $G \times E$

Ultimate Guide to Understand Engine Behind $G \times E$

(Dedicated to IRRI's Legacy!)



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Introduction

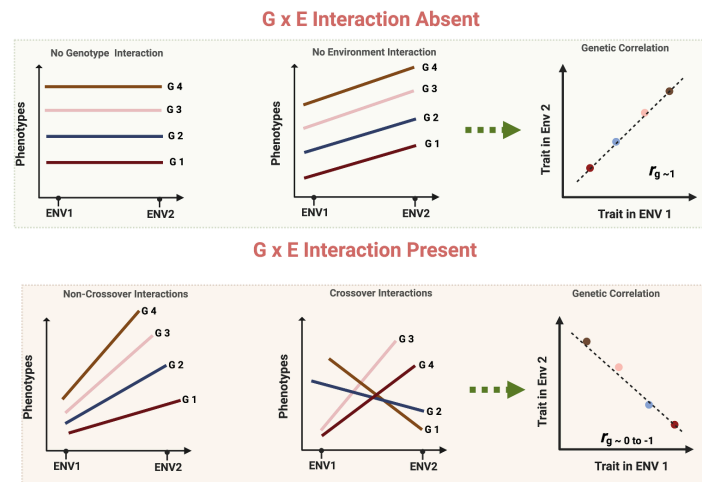
Welcome back, brave traveler of statistical realms! You've already survived [Henderson's Mixed Model Equations](#), wrestled with BLUEs and BLUPs, and even hand-computed a few before realizing that such heroics are best left to computers.

And now—we face the *great dragon of quantitative genetics: Genotype-by-Environment ($G \times E$) interactions*.

Why does **Genotype A** strut around like a rockstar in one location, only to flop harder than a soggy sandwich in another? And why is **Genotype B** the underdog in one environment but suddenly the Beyoncé of the field trial somewhere else? That, in a nutshell, is the rollercoaster called **$G \times E$ interactions** in crop breeding. The figure pretty much spills the tea all by itself.

What is $G \times E$ Environment Interaction

- **Phenotypic Plasticity:** a changing the phenotypic expression due to change in environment
- Phenotypic plasticity: The range of phenotypes a single genotype can express as a function of its environment.
- When phenotypic plasticity differs between genotypes, this is described as **$G \times E$**
- $G \times E$ is commonly defined as the joint effect of genetics and the environment that deviates from their individual additive effects



Created in BioRender. Hussain, W. (2024)
<https://BioRender.com/v23p670>

In this tutorial we unroll the magical scrolls to understand the driving engine behind the $G \times E$ interactions and demystify the $G \times E$ interactions

Strap in—our quest begins with **Henderson's Mixed Model Equations (MME)**, this time wielded against the chaos of $G \times E$.

We will carefully build the model step by step: **exploring the structure of $G \times E$** , revealing the MME formulation with genotypes and environments as random effects, replications as fixed, and $G \times E$ as random. Together, we'll assemble the matrices, **align the Left-Hand Side (LHS) against the Right-Hand Side (RHS)**, and finally solve for BLUEs and BLUPs—this time in the **realm of $G \times E$ interactions**.

But the adventure doesn't end there. As done in previous tutorial, We will march further, comparing our **hand-crafted MME solutions** against the predictions of ***lme4 R Package***. Will the rankings match between our manual matrix magic and what the software conjures? Only one way to find out.

So, get ready—fire up your statistical swords. First, let's summon our assistants—not the kind who fetch coffee, but the R functions that channel mixed model wizardry:

```
> # Load the library
> library(lme4)
```

Data Information

Get your backpacks ready—our quest uses a real dataset from *IRRI (International Rice Research Institute)*. This *humble .csv*, tucked deep in the archives, contains the stuff of legend: **2 environments**, **200 genotypes** braving the trials with **2 replications** (because every hero deserves a second chance).

Our mission: Dissect $G \times E$ interactions using Yield as our Treasure Map.

Here's how we'll wield our powers: - Fixed effect: Replication (2 levels; because life—and statistics—deserves a re-run) - Random effects: Genotypes (200 levels; a stadium full of competitors) - Environments: (2; the classic “Desert vs Jungle” scenario) - $G \times E$ interaction: ($200 \times 2 = 400$ possible combinations! A rollercoaster of drama.)

And yes — a sprinkle of humor will make the journey more epic.

Let's awaken our dataset in R (no need for magic spells, just tidy code):

```
> rm(list=ls()) # Remove previous work
> # Load data
> demo.data <- read.csv("./Data/demo.data1.csv", header = TRUE)
> # Factor conversion
> demo.data$Environment <- as.factor(demo.data$Environment)
> demo.data$Rep <- as.factor(demo.data$Rep)
> demo.data$Genotype <- as.factor(demo.data$Genotype)
> # Interaction term for GxE
> demo.data$gen_env <- interaction(demo.data$Genotype, demo.data$Environment)
> # str(demo.data) makes sure everything's in order-no stray variables
> str(demo.data)
'data.frame': 800 obs. of 10 variables:
 $ Environment: Factor w/ 2 levels "Env1","Env2": 1 1 1 1 1 1 1 1 1 1 ...
 $ Plot       : int 59 19 27 127 132 92 106 82 199 57 ...
 $ Genotype   : Factor w/ 200 levels "1","2","3","4",...: 1 2 3 4 5 6 7 8 9 10 ...
 $ Rep        : Factor w/ 2 levels "1","2": 1 1 1 1 1 1 1 1 1 1 ...
 $ Block      : int 2 1 1 4 4 3 3 3 5 2 ...
 $ Row        : int 2 1 1 4 4 3 3 3 5 2 ...
 $ Column     : int 19 19 27 7 12 12 26 2 39 17 ...
 $ Line.type  : chr "Entry" "Entry" "Entry" "Entry" ...
 $ Yield      : num 5873 5844 5777 6284 5703 ...
 $ gen_env    : Factor w/ 400 levels "1.Env1","2.Env1",...: 1 2 3 4 5 6 7 8 9 10 ...
```

Show Data as Table

```
> kable(head(demo.data))
```

Environment	Plot	Genotype	Rep	Block	Row	Column	Line.type	Yield	gen_env
Env1	59	1	1	2	2	19	Entry	5873.292	1.Env1
Env1	19	2	1	1	1	19	Entry	5843.605	2.Env1
Env1	27	3	1	1	1	27	Entry	5776.744	3.Env1
Env1	127	4	1	4	4	7	Entry	6283.830	4.Env1
Env1	132	5	1	4	4	12	Entry	5703.488	5.Env1
Env1	92	6	1	3	3	12	Entry	6357.558	6.Env1

Model Description

Now that our data party is assembled, let's introduce the main event: the **Mixed Model**—a statistical festival where **fixed effects** and **random effects** mingle, dance, and sometimes trip over each other's code.

When the quest involves **random effects** for **Genotypes**, **Environments**, and their magical **interaction (G × E)**. The **Replication** as the wise **fixed Effects**. And our ancient spellbook—**Henderson's Mixed Model Equations**—evolves to handle all these Effects.

MME with Interactions

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}_g\mathbf{u}_g + \mathbf{Z}_e\mathbf{u}_e + \mathbf{Z}_{ge}\mathbf{u}_{ge} + \boldsymbol{\varepsilon}$$

Y is the Yield

$\mathbf{X}$ is design matrix relating phenotypes to Replications
 $\boldsymbol{\beta}$ is a vector of fixed effects (Replications)

$\mathbf{Z}_g$ is a design matrix of random effects of Genotypes
 $\mathbf{u}_g$ is a vector of random effects with variance= $\text{Var}(\mathbf{u}) \sim \sigma_g^2 \mathbf{I}$, where $\mathbf{I}$ is identify matrix (no-covariance structure)

$\mathbf{Z}_e$ is a design matrix of random effects of Environments
 $\mathbf{u}_e$ is a vector of random effects of environments with variance= $\text{Var}(\mathbf{u}_e) \sim \sigma_e^2 \mathbf{I}$, where $\mathbf{I}$ is identify matrix (no-covariance structure: see on right side for various models)

$\mathbf{Z}_{ge}$ is a design matrix of G x E interactions.
 $\mathbf{u}_{ge}$ is a vector of random effects with variance= $\text{Var}(\mathbf{u}_{ge}) \sim \sigma_{ge}^2 \mathbf{I}$, where $\mathbf{I}$ is identify matrix (no-covariance structure)

$\boldsymbol{\varepsilon}$ is $n \times n$ matrix of residual/error effects with $\text{Var}(\boldsymbol{\varepsilon}) = \sigma_e^2 \mathbf{I}$, $\mathbf{I}$ is identity matrix.

Hussain et al. 2024 (BMC Plant Methods)

- **Model 1 and model 2:** same variance (homogeneous)
- **Model 3:** Different variances across environments (heterogeneous)
- **Model 4:** Same as Model 3 with Spatial variations.
- **Model 5:** Spatial variations specific to each environment.
- **Model 6:** Heterogeneous in variance but no correlation between environments
- **Model 7:** Uniform correlation between environments
- **Model 8 and 9:** Factor analytical models with k multiplicative components
- **Model 10:** Factor analytical model with Spatial variations specific to each environment.

The grand formula for our mixed model saga in detail is given below. Here our Model is simple (Model1), no assumption of co-variance structures. See on rightside of figure above.

$$y = X\beta + Z_g u_g + Z_e u_e + Z_{ge} u_{ge} + \varepsilon \quad (1)$$

where,

y : is the vector of Yield
 X : is the incidence matrix for fixed effects-replications
 β : is the fixed effect of replication
 Z_g : is the incidence matrix for random Genotypes
 u_g : is the random effect of genotypes
 Z_e : is the incidence matrix for random Environments
 u_e : is the random effect of environments
 Z_{ge} : is the incidence matrix for random G x E effects
 u_{ge} : is the random effect G x E effects
 ε : is residuals

here,

$$\begin{aligned}u_g &\sim N(0, I\sigma_{u_g}^2) \\u_e &\sim N(0, I\sigma_{u_e}^2) \\u_{ge} &\sim N(0, I\sigma_{u_{ge}}^2) \\\epsilon &\sim N(0, I\sigma_{\epsilon}^2)\end{aligned}$$

In this quest, we're keeping life simple: **there is no covariance either among genotypes, across environments, or even between interactions—each gets its own magical forcefield** (the identity matrix I), and their fates are statistically independent, wandering the land solo without forming mischievous alliances.

This means: **Genotypes** do their thing, **environments** do theirs, and **G × E** interactions party alone. No side-eye glances, shared secrets, or underground tunnels connecting the variance structures. In technical speak: **independent random effects and homogenous variances**.

For now, our shields are simple and mighty. More complex models—those with elaborate covariance between genotypes (think pedigrees), or environments (think spatial correlation), or even diagonal/block structures—are advanced topics for another guild meeting.

The MME Structure

Think **MME with G x E interactions** as a grand dinner party where genotypes, environments, and replications all politely keep their distance and focus on their own plates. And Imagine we want to **predict yield and get BLUEs and BLUPs** using:

- **Fixed effects** (replication, environment, etc.)
- **Random effects** for genotypes, environments, and G × E interaction.

We set up one big matrix equation like assembling four rows of ingredients for a delicious cake—each row and column tells you which ingredient is mixed with which!

Presenting the Legendary MME Structure

$$\begin{pmatrix} X'X & X'Z_g & X'Z_e & X'Z_{ge} \\ Z_g'X & Z_g'Z_g + \frac{1}{\sigma_g^2}I & Z_g'Z_e & Z_g'Z_{ge} \\ Z_e'X & Z_e'Z_g & Z_e'Z_e + \frac{1}{\sigma_e^2}I & Z_e'Z_{ge} \\ Z_{ge}'X & Z_{ge}'Z_g & Z_{ge}'Z_e & Z_{ge}'Z_{ge} + \frac{1}{\sigma_{ge}^2}I \end{pmatrix} \begin{pmatrix} \hat{\beta} \\ \hat{u}_g \\ \hat{u}_e \\ \hat{u}_{ge} \end{pmatrix} = \begin{pmatrix} X'y \\ Z_g'y \\ Z_e'y \\ Z_{ge}'y \end{pmatrix}$$

Look around—there are no mysterious G covariance matrices in Structure, It is replaced by I in MME structure. That means each genotype has its own diagonal variance and fiercely independent (No relationship)! You won't find genotypes whispering secrets to environments or interactions; all cross-talk stays mathematical and professional.

Breaking Down Each Part (Like a Recipe!)

Rows and columns: Each row and column represents the connections between an effect (fixed, genotype, environment, G×E) and all the others. - Each element is a cross-product that matches up how these effects appear in your data.

Diagonal Blocks:

- The diagonal terms (like

$$Z_g'Z_g + \frac{1}{\sigma_g^2}I$$

) mix the “self-effect” for each random component with an extra dash—the shrinkage (variance) for that effect. I is identify matrix! No G .

- This makes sure random effects don't get too wild!

Off-diagonal Blocks:

- The off-diagonals (like

$$X'Z_g, Z_g'Z_e$$

) measure how the different effects interact with each other.

Left-hand side (LHS): The engine room. Here, all effects combine their powers to solve the equations and generate heroic estimates.

Right-hand side (RHS): The command center. Where the data's wisdom gets handed off for swift computation and interpretation.

Who's Who at the Table?

- The $X'X$ block is Replication flexing its fixed-effect muscles—all about consistency!
- Those blocks with $\frac{1}{\sigma_g^2}I$, $\frac{1}{\sigma_e^2}I$, and $\frac{1}{\sigma_{ge}^2}I$ are the variance captains—armed with solo shields, not looking for trouble.
- Off-diagonal blocks record the polite exchanges: “Did you see Genotype waving at Environment?” Just a nod, no handshakes.

This is “basic $G \times E$ fortress”—no leaks, no subplots, totally drama-free. All random effects are independent, all shields are up. The fancy covariance dragons, diagonal dungeons, and intricate genetic entanglements? Save those for your next-level quest—this is your training ground.

When you finally crack this matrix stronghold and solve for $\hat{\beta}$, $\hat{u}_g$, $\hat{u}_e$, and $\hat{u}_{ge}$, you'll hold the magical script for predicting yields, ranking genotypes, and winning statistical glory—with matrix heroism at its boldest and cleanest!

Summary of MME structure

We **fill in each block** with your design matrices (**X for fixed, Z for random**), sprinkle in known variances to control **shrinkage**, and stack everything in the right spot. We will **Solve** the big system and out pop your fixed effects (BLUES) and random effects (BLUPs)—the recipe for genetic insights! If you change the model (like leaving out $G \times E$), you just drop that row and column—like leaving out an ingredient like a basic MME showed last time!

Build Incidence/Design Matrices

Now, the real fun begins: welcome to the Matrix Building Olympics! Grab your mathematical sweatbands, because it's time to flex those R skills and stack some serious statistical bricks.

Picture your dataset like a LEGO empire. Each incidence matrix is a different color block—designed to fit together in mixed model glory. With every line of R code, you assemble your team, multiplying your way to MME greatness.

Here's how to build this superhero squad in R:

```
> # Incidence/design matrix for fixed effects: Rep
> X <- model.matrix(~ Rep, demo.data)
> # Random effect design matrix for Genotype
> Z_g <- model.matrix(~ -1 + Genotype, demo.data)
> # Random effect design matrix for Environment
> Z_e <- model.matrix(~ -1 + Environment, demo.data)
> # Random effect design matrix for G x E
> Z_ge <- model.matrix(~ -1 + gen_env, demo.data)
```

Multiply, stack, and prepare—soon these matrices will form the mighty MME block structure, ready for statistical battle. Each matrix is an independent warrior: no cozying up, just pure algebraic action.

Variance Components

Wait—before you build your statistical fortress, don’t forget to mix up your **secret sauce**: the variance components! As described in last tutorial, normally, these magical numbers are brewed in a cauldron of high-tech computation using **REML (Restricted Maximum Likelihood)**—a process so intense it makes *lme4* sweat and computers hum.

But since we’re taking the hands-on, “grandma’s kitchen” approach, there’s no waiting around for algorithms: **we simply plug in our variance components, freshly estimated by lme4 and delivered on a shiny platter**. Picture it:

- Instead of the kitchen drama of REML, we’re just grabbing the ingredients and tossing them straight into the recipe.
- I extracted these variance components from running `VarCorr(mod_gxe_env)` in *lme4* (or similar).

Here’s your plug-and-play recipe in R and the **shrinkage factors** (

$$\lambda_g, \lambda_e, \lambda_{ge}$$

) in your code calculated as follows:

```
> # Enter your estimated variance components, usually from lme4::VarCorr(mod_gxe_env)
> # Genotype variance
> sigmau <- 114010
> # Residual variance
> sigmae <- 331603
> # Environment variance
> sigmaenv <- 16218
> # GxE variance
> sigmage <- 229771
> # Build Diagonal Matrix for each: assumption that all random effects are independent and have equal va
> I_g <- diag(ncol(Z_g))
> I_e <- diag(ncol(Z_e))
> I_ge <- diag(ncol(Z_ge))
> # Calculate lambda (variance ratios): Shrinkage factors.
> lambda_g <- sigmae / sigmau # For genotypes
> lambda_e <- sigmae / sigmaenv # For environments
> lambda_ge <- sigmae / sigmage # For g x e Interactions
```

Lambda values turn your variance estimates into squaring-off ratios for the MME diagonal blocks—essential seasoning for matrix algebra. We’ll use these to “enhance” our diagonal blocks in the MME structure. It’s fast, fun, and—most importantly—lets everyone get to the good part (solving!) without a pit stop in Particle Physics Computing Lab.

Matrix Multiplication: Cross Products

Here comes the part where statistics feels a bit like bodybuilding—crunching out cross-products to bulk up your MME block matrix! Step by step, you’ll flex each pairing of your fixed and random effect matrices, matching them perfectly to every block in the MME fortress.

Each matrix multiplication is a handshake at the banquet: sometimes a dramatic high-five (diagonal blocks), sometimes a polite nod from across the table (off-diagonal blocks).

Here’s how your cast of matrices lines up with the MME structure above:

Fixed × Fixed

This is $X'X$ (X transpose X): Replication talking to itself. The fixed effect is checking its own reflection.


```
> XtX <- t(X) %*% X
```

Fixed × Random

Each is $X'Z$ type, representing Rep's attempt to reach out to the Genotype, Environment, and $G \times E$ squads.

```
> XtZg <- t(X) %*% Z_g # (X'Z_g) of MME
> XtZe <- t(X) %*% Z_e # (X'Z_e) of MME
> XtZge <- t(X) %*% Z_ge # (X'Z_{ge}) of MME
```

Random × Fixed

Flip side: random effects waving back at Replication. Courtesy goes both ways!

```
> ZgtX <- t(Z_g) %*% X # (Z_g'X) of MME
> ZetX <- t(Z_e) %*% X # (Z_e'X) of MME
> ZgetX <- t(Z_ge) %*% X # (Z_{ge}'X) of MME
```

Random × Random

Here's where all genotypes, environments, and their interactions get to size each other up—or simply check their own squad's strength on the diagonal ($Z'Z + \text{varianceterm}$), and gossip about each other in off-diagonals.

```
> ZgtZg <- t(Z_g) %*% Z_g # (Z_g'Z_g) of MME
> ZgtZe <- t(Z_g) %*% Z_e # (Z_g'Z_e) of MME
> ZgtZge <- t(Z_g) %*% Z_ge # (Z_g'Z_{ge}) of MME
>
> ZetZg <- t(Z_e) %*% Z_g # (Z_e'Z_g) of MME
> ZetZe <- t(Z_e) %*% Z_e # (Z_e'Z_e) of MME
> ZetZge <- t(Z_e) %*% Z_ge # (Z_e'Z_{ge}) of MME
>
> ZgetZg <- t(Z_ge) %*% Z_g # (Z_{ge}'Z_g) of MME
> ZgetZe <- t(Z_ge) %*% Z_e # (Z_{ge}'Z_e) of MME
> ZgetZge <- t(Z_ge) %*% Z_ge # (Z_{ge}'Z_{ge}) of MME
```

How These Fit the MME Fortress

Every block from your mighty MME matrix connects directly to these cross-products: - Diagonal blocks are all about self-respect: each group pats itself on the back, shielding itself in battle. - Off-diagonal blocks are the statistical “hello’s” between squads—measurements of overlap and teamwork (or lack thereof).

And with each multiplication, your model grows stronger, clearer, and more ready to face the forces of $G \times E$! Each block lands perfectly in its MME slot, exactly as in the fortress blueprint.

Assemble Left-Hand Side

Now time to play a game of assembling those cross-product blocks into the **legendary MME mega-structure** on LHS! Every chunk gets slotted into position, building the mathematical fortress that will guard our BLUEs and BLUPs. Let's break it down step by epic step:

Build the Four Rows of Destiny Imagine this like prepping a sandwich—each layer has a special ingredient:

LHS_top (The Bun):

- The fixed effect squad takes the lead, joined by all random effect warriors for a hearty start.
- Matches the first row in our MME fortress—Replication surrounded by everyone else.

```
> LHS_top <- cbind(XtX, XtZg, XtZe, XtZge)
```

LHS_mid1 (The Protein):

- Genotypes flex their muscles, protected by their variance shields ($\sigma^2 + \lambda_g I_g$).
- This is the Genotype row, always keeping things mysterious and deliciously random.

```
> LHS_mid1 <- cbind(ZgtX, ZgtZg + lambda_g * I_g, ZgtZe, ZgtZge)
```

LHS_mid2 (The Cheese):

- Environments chime in next, bold and unpredictable, buoyed by their own variance armor.
- Sits tight in the third row of the fortress, gleaming and slightly pungent (as environments sometimes are).

```
> LHS_mid2 <- cbind(ZetX, ZetZg, ZetZe + lambda_e * I_e, ZetZge)
```

LHS_bot (The Secret Sauce):

- Genotype $\times$ Environment interactions swirl in for the final flavor, bringing complexity and pizzazz!
- The fourth row, ready to tie all the ingredients together like culinary magic.

```
> LHS_bot <- cbind(ZgetX, ZgetZg, ZgetZe, ZgetZge + lambda_ge * I_ge)
```

Step 2: Stack the Fortress Layer up those rows! Just like assembling a heroic party in an RPG, everyone gets their rightful place:

- This is your full MME block structure—the **Left-Hand Side (LHS) of our epic equation!**
- Each block maps to one line in the MME matrix shown earlier: Replication, Genotype, Environment, and $G \times E$ interaction—all flexing their stats loud and proud.

```
> LHS <- rbind(LHS_top, LHS_mid1, LHS_mid2, LHS_bot)
```

The RHS—The Council of MME

On the other side of destiny sits the **Right-Hand Side (RHS)**, holding the wisdom and secrets of actual field data: Each chunk represents a council member's verdict on Yield, forming the full RHS to match the LHS.

```
> # Replication's view of Yield
> Xty <- t(X) %*% demo.data$Yield
> # Genotype's take
> Zgty <- t(Z_g) %*% demo.data$Yield
> # Environment's feedback
> Zety <- t(Z_e) %*% demo.data$Yield
> # G x E interaction's drama
```

```
> Zgety <- t(Z_ge) %*% demo.data$Yield
> # RHS
> RHS <- rbind(Xty, Zgty, Zety, Zgety)
```

Marvel at the Structure

Just as foretold, every block in our fortress aligns perfectly with the original MME blueprint: welcoming fixed effects, random effects, and interactions. The *shrinkage factors*—like magical shields—make sure the random effects don’t get too wild, gently coaxing everything toward a sensible mean.

So, what happens next? By solving

$$\text{LHS} \times \text{solution vector} = \text{RHS}$$

you unleash the true might of **BLUES** (Best Linear Unbiased Estimates, for your fixed effects like Replication) and **BLUPs** (Best Linear Unbiased Predictions, for those unpredictable genotypes, environments, and $G \times E$ combos).

It’s more than matrix multiplication—think of assembling the statistical Avengers for $G \times E$ greatness, each hero popping out with the best possible estimate or prediction—no bias, minimum variance, and plenty of teamwork.

Solve the Extended MME

Here finally comes your magical command in R:

```
> sol <- solve(LHS, RHS)
```

In one line, you crack open the vault of the mixed model, summoning all your BLUES and BLUPs for the field trial epic!

The solution vector contains **BLUES for each fixed effect** and **BLUPs for each random term**. Whether it’s a top-performing genotype or a mysterious environment, *cling tight* to your solution vector—your journey into $G \times E$ just found its treasure!

Extract BLUES and BLUPs

After solving the mighty MME system, the **solution vector** is like Pandora’s box—crack it open, and you’ll find all the secrets of your trial tucked in neat order: Here we will extract the BLUES and BLUPs.

The **BLUES** are the first chunk of your solution; these are your fixed effects, wearing the crown of “Best Linear Unbiased Estimates.” They’re the wise elders (like overall mean and replicates) who always speak for the group. The **BLUPs for genotypes** immediately follow; they’re the unpredictable rock stars, each getting the “Best Linear Unbiased Prediction” based on both their personal performance and the wider field trial story. The **BLUPs for $G \times E$ interactions** come next—these wild cards show how each genotype-environment combo dances differently depending on where you plant it.

View Sol containing fixed and random effects for all as table

```
> # Saving the sol in Outputs as dataframe
> Outputs<-data.frame(sol)
> kable(head(Outputs))
```

	sol
(Intercept)	5196.42171
Rep2	-60.27981
Genotype1	97.30569
Genotype2	-146.08324
Genotype3	-12.74910
Genotype4	457.02296

```

> # Number of fixed effects (Replication)
> n_fixed <- ncol(X)
> # Number of genotypes is 200
> n_gen <- ncol(Z_g)
> # Number of environments is 2
> n_env <- ncol(Z_e)
> # Number of genotype x environment 400
> n_ge <- ncol(Z_ge)
> # BLUEs for Replication effects (Second)
> BLUEs <- sol[1:n_fixed, ]
> # BLUPs for genotype effects (starts after Blues of replications)
> BLUPs_gen <- sol[(n_fixed+1):(n_fixed+n_gen), ]
> # BLUPs for environment (starts after replication and genotypes effects)
> BLUPs_env <- sol[(n_fixed+n_gen+1):(n_fixed+n_gen+n_env), ]
> # BLUPs for g x e interaction effects (starts after fixed effect plus genotype plus environment)
> BLUPs_ge <- sol[(n_fixed+n_gen+n_env+1):(n_fixed+n_gen+n_env+n_ge), ]
> # Add intercept for comparison (assume first BLUE is intercept)

```

View all as Table

```

> intercept <- BLUEs[1]
> BLUPs_G_MAN <- data.frame(BLUPsG_MAN=BLUPs_gen + intercept) # BLUPs for Genotypes
> kable(head(BLUPs_G_MAN))

```

	BLUPsG_MAN
Genotype1	5293.727
Genotype2	5050.338
Genotype3	5183.673
Genotype4	5653.445
Genotype5	5229.727
Genotype6	5277.335

```

> BLUPs_Env_MAN <- data.frame(BLUPsE_MAN=BLUPs_env + intercept) # BLUPs for Environment
> kable(head( BLUPs_Env_MAN))

```

	BLUPsE_MAN
EnvironmentEnv1	5281.438
EnvironmentEnv2	5111.405

```

> BLUPs_GE_MAN <- data.frame(BLUPsGE_MAN=BLUPs_ge + intercept)# BLUPs for G X e
> kable(head(BLUPs_GE_MAN ))

```

	BLUPsGE_MAN
gen_env1.Env1	5247.561
gen_env2.Env1	5423.371
gen_env3.Env1	5307.931
gen_env4.Env1	5781.357
gen_env5.Env1	5271.724
gen_env6.Env1	5498.846

Additional Work

This script below is additional work to make sure latter when we compare with lme4 the data frames are in order to match. I am just creating 2 columns and using it to arrange so that BLUPs are grouped and sorted so each genotype appears sequentially across both environments (This is the output we get in lme4)

```
> BLUPs_GE_MAN$Genotype <- sub("\\.Env[12]$", "", rownames(BLUPs_GE_MAN))
> BLUPs_GE_MAN$Environment <- sub(".*\\. (Env[12])$", "\\1", rownames(BLUPs_GE_MAN))
> BLUPs_GE_MAN_ordered <- BLUPs_GE_MAN[order(BLUPs_GE_MAN$Genotype, BLUPs_GE_MAN$Environment), ]
> BLUPs_GE_MAN_ordered$Genotype<- sub("^gen_env", "", BLUPs_GE_MAN_ordered$Genotype)
> BLUPs_GE_MAN_ordered <- BLUPs_GE_MAN_ordered[order(as.numeric(BLUPs_GE_MAN_ordered$Genotype)), ]
```

Here's a summary paragraph, written in a friendly and intriguing style, to tell users how solving MME really works:

This is how the magic of Mixed Model Equations (MME) unfolds! Imagine assembling a puzzle, where each piece is a block of your data—fixed effects, random effects, and those wild interaction terms—all coming together in one big matrix recipe. The secret sauce? Matrix multiplication. You simply stack these cross-products, stir in your variance ingredients, and let the system work its wonders

No mystery, no smoke and mirrors: when any fancy software gives you BLUEs and BLUPs, you can now nod knowingly—inside, it's just this grand matrix equation being solved, piece by piece. Solving the MME is truly a cakewalk once you see how it all fits together!

Run Analysis in lme4

Now comes the “Hollywood remake” of your manual MME saga—using lme4’s statistical special effects! Just as in your homemade matrix heroics, the computerized lmer() will estimate all the same BLUEs and BLUPs for the fixed and random effects, including genotypes, environments, and $G \times E$ interactions.

Here's your lme4 code:

This model specifies:

- env as a fixed effect (can be random if you remove as a fixed) - (1 | gen) for genotypes (BLUPs!),
- (1 | env) for environments (BLUPs!),
- (1 | gen:env) for $G \times E$ interaction (BLUPs!)

```
> # G x E Model in lme4
> mod_gxe <- lmer(Yield ~ Rep + (1 | Genotype) + (1 | Environment) + (1 | Genotype:Environment),
+               data = demo.data)
> # Get Summary of model
> summary(mod_gxe)
```

```

Linear mixed model fit by REML ['lmerMod']
Formula:
Yield ~ Rep + (1 | Genotype) + (1 | Environment) + (1 | Genotype:Environment)
Data: demo.data

REML criterion at convergence: 12860.3

Scaled residuals:
    Min       1Q   Median       3Q      Max
-2.7698 -0.5494 -0.0175  0.5418  4.4784

Random effects:
Groups                Name      Variance Std.Dev.
Genotype:Environment (Intercept) 229771  479.3
Genotype              (Intercept) 114010  337.7
Environment           (Intercept)  16218  127.4
Residual                          331603  575.9
Number of obs: 800, groups:
Genotype:Environment, 400; Genotype, 200; Environment, 2

Fixed effects:
              Estimate Std. Error t value
(Intercept)  5196.42     100.41   51.75
Rep2          -60.28      40.72   -1.48

Correlation of Fixed Effects:
      (Intr)
Rep2 -0.203

```

How to Extract Estimates

BLUEs (fixed effects: Intercept plus Replication):

```

> # For Replications
> fixef_vals <- fixef(mod_gxe)
> fixef_vals
(Intercept)      Rep2
 5196.42171   -60.27982

```

BLUPs (random effects of Genotypes)

```

> # Get BLUPs for Genotypes and add intercept
> BLUP_G_lme4 <- data.frame(BLUPsG_LME4=ranef(mod_gxe)$Genotype[,1] + fixef_vals[1])
> # Add names similar to manually generated file
> row.names(BLUP_G_lme4)<-paste0("Genotype", row.names(BLUP_G_lme4))
> # Get BLUPs for Environment and add intercept
> BLUP_E_lme4 <- data.frame(BLUPsE_LME4=ranef(mod_gxe)$Environment[,1] + fixef_vals[1])
> # Add names similar to manually generated file
> row.names(BLUP_E_lme4)<-paste0("EnvironmentEnv", row.names(BLUP_E_lme4))
> # Get BLUPs for G x E and add intercept
> BLUP_GE_lme4 <- data.frame(BLUPsGE_LME4=ranef(mod_gxe)$`Genotype:Environment`[,1] + fixef_vals[1])

```

To ensure a **fair comparison** between the manual MME solution and lme4's results, it's important to add the model's intercept (and any fixed effects like environment means if relevant) to all BLUPs for genotypes, environments, and G

× E interactions. This gives **predicted values** on the same scale, combining the overall average yield (the intercept or fixed effect) with the specific effect for each level.

Here's how you do it in R:

```
> # Extract the fixed effects (includes the intercept)
> fixef_vals <- fixef(mod_gxe)
> # BLUPs for genotypes
> blup_gen <- ranef(mod_gxe)$gen[,1] + fixef_vals[1]
> # BLUPs for environments
> blup_env <- ranef(mod_gxe)$env[,1] + fixef_vals[1]
> # BLUPs for G × E interaction
> blup_gxe <- ranef(mod_gxe)$`gen:env`[,1] + fixef_vals[1]
```

Compare Manual vs. lme4 Results

Now, having both sets of estimates, we can: **Line up the manual MME solution vector against the lme4 BLUPs and BLUEs.**

We will do a side-by-side face-off! We will round the values first and check whether two are equal.

```
> # BLUPs of Genotypes
> table(round(BLUPs_G_MAN$BLUPsG_MAN, 2) == round(BLUP_G_lme4$BLUPsG_LME4, 2))

TRUE
200
> # BLUPs of G x E
> table(round(BLUPs_GE_MAN_ordered$BLUPsGE_MAN, 1) == round(BLUP_GE_lme4$BLUPsGE_LME4, 1))

FALSE TRUE
1 399
```

The tension was high as the two titans—**Manual BLUPs** and **lme4 BLUPs**—went head-to-head in a dramatic, side-by-side showdown! With trembling fingers and bated breath, we rounded their values for a true apples-to-apples comparison. **The verdict? All TRUE!** Every genotype's BLUP matched perfectly, like synchronized swimmers in a pool of data. Cue the confetti, fire up the cake, and let the coffee mugs clink—statisticians everywhere rejoice!

But wait—one solitary “FALSE” snuck through in the G × E ranks. Was it a glitch in the universe? A cosmic decimal point rebellion? That is it!

Comapre the files

```
> BLUPs_G_COMB <- cbind(BLUPs_G_MAN, BLUP_G_lme4)
> kable(head(BLUPs_G_COMB))
```

	BLUPsG_MAN	BLUPsG_LME4
Genotype1	5293.727	5293.727
Genotype2	5050.338	5050.338
Genotype3	5183.673	5183.673
Genotype4	5653.445	5653.445
Genotype5	5229.727	5229.727
Genotype6	5277.335	5277.335

```
> row.names(BLUP_GE_lme4) <- row.names(BLUPs_GE_MAN_ordered)
> BLUPs_GE_COMB <- cbind(BLUPs_GE_MAN_ordered, BLUP_GE_lme4)
> kable(head(BLUPs_GE_COMB))
```

	BLUPsGE_MAN	Genotype	Environment	BLUPsGE_LME4
gen_env1.Env1	5247.561	1	Env1	5247.561
gen_env1.Env2	5341.388	1	Env2	5341.388
gen_env2.Env1	5423.371	2	Env1	5423.371
gen_env2.Env2	4675.062	2	Env2	4675.062
gen_env3.Env1	5307.931	3	Env1	5307.931
gen_env3.Env2	5059.218	3	Env2	5059.218

Time to celebrate: we’ve conquered the matrix and the model, with only decimal dust left behind. □□

Conclusion

The lme4 package is like your dependable research assistant who doesn’t need coffee breaks—it speeds through the same calculations and hands you BLUEs and BLUPs as neatly as if they’d been gift-wrapped. Comparing it with “**old-school matrix magic**” shows that both ultimately open the same mixed-model treasure chest, just via different routes: one through manual algebraic hiking, the other on an express statistical bullet train.

The purpose here is simple: to reveal where those *mysterious numbers* actually come from, making the math less like sorcery and more like well-understood science—with just a dash of sparkle left in for fun.

Additional Reading

Here’s a treasure trove of references for further reading—because every tutorial needs some “bonus features.” Dive into these materials if you want to keep the science party going (and maybe impress friends at dinner with plant genetics facts). Enjoy the tutorial... and remember, real scientists always check the references (preferably with snacks nearby).

- [Linear Mixed-Effects Models in chemistry: A tutorial](#)
- [Shrinkage in Nonlinear Mixed-Effects Population Models: Quantification, Influencing Factors, and Impact](#)
- [Shrinkage in Nonlinear Mixed-Effects Population Models: Quantification, Influencing Factors, and Impact](#)
- [Open-source analytical pipeline for robust data analysis, visualizations and sharing in crop breeding](#)
- [The statistical analysis of multi-environment data: modeling genotype-by-environment interaction and its genetic basis](#)
- [From Genotype × Environment Interaction to Gene × Environment Interaction](#)
- [Genotype by Environment analysis using statgenGxE](#)
- [Towards understanding and use of mixed-model analysis of agricultural experiments](#)
- [Linear Mixed Effects Models in R](#)
- [Mixed Models: Testing Significance of Effects](#)
- [An Introduction to Linear Mixed Models in R and SAS](#)

For any suggestions or comments, please feel to reach at waseem.hussain@cgiar.org